How a Distractor on One Hand Changes Judgments on the Other

Research update

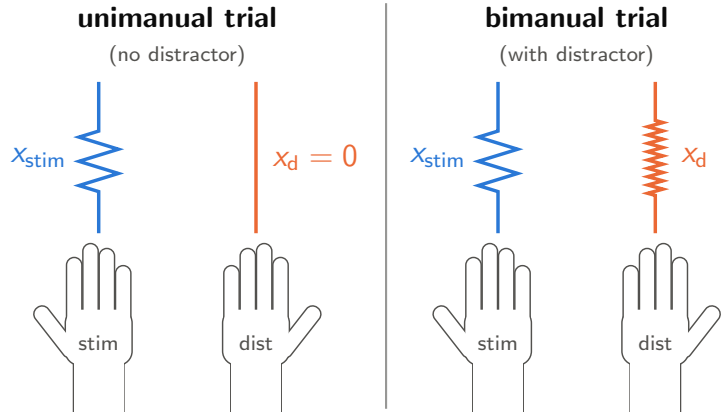
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Today

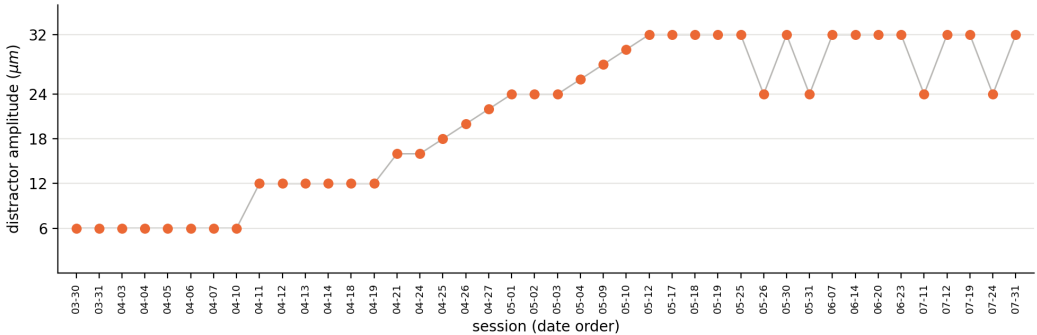
1. The terms I use for the data
2. Three models of the data
3. What they show
4. Open questions for you

Two kinds of trial



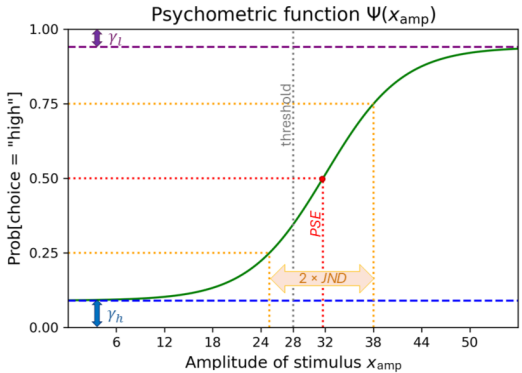
Group: hand $\times$ trial type, so four groups: left/right, unimanual/bimanual

Each session has one distractor amplitude



42 sessions of about 300 trials, from one monkey (Sirius)

The psychometric curve



- **PSE** (bias): called "high" half the time
  lower PSE = more "high" answers
- **JND** (sensitivity): how fast the curve rises
  larger JND = less precise
- **Lapses:** answers the stimulus does not explain
  γ_h = floor, γ_l = gap below the ceiling

Distractor effect = bimanual − unimanual, for each hand

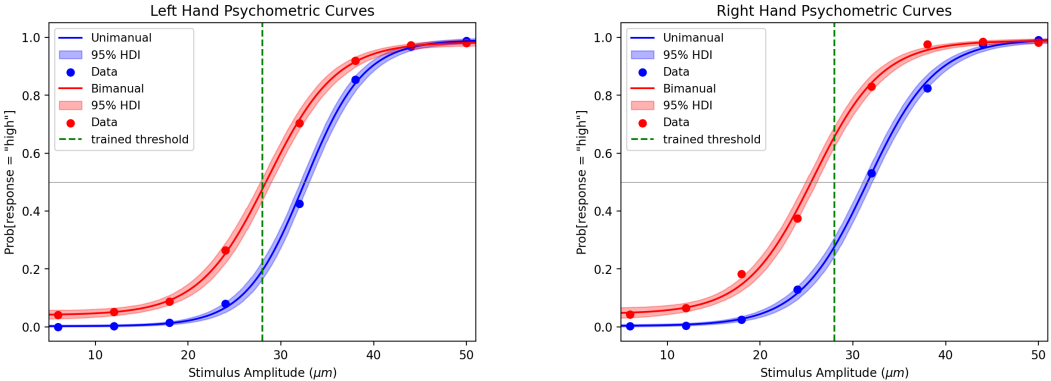
Three models

- **Model A:** one curve per group, all sessions pooled
- **Model B:** a curve for every session
  sessions may differ, but are assumed to resemble each other
- **Model C:** like B, but the distractor's shift is proportional to its amplitude
  one curve per hand, with both trial types

How I check a model

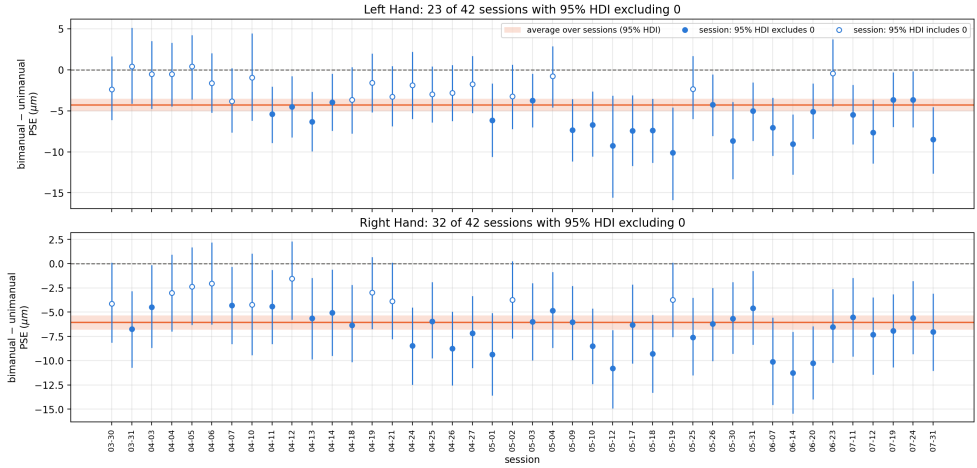
1. **Simulated data:** make data with known answers, and check that the model gets them back
2. **Real data:** check that the fitted model reproduces what the monkey did
3. **Compare models:** how well each predicts trials left out of its fit

The distractor makes “high” answers more likely



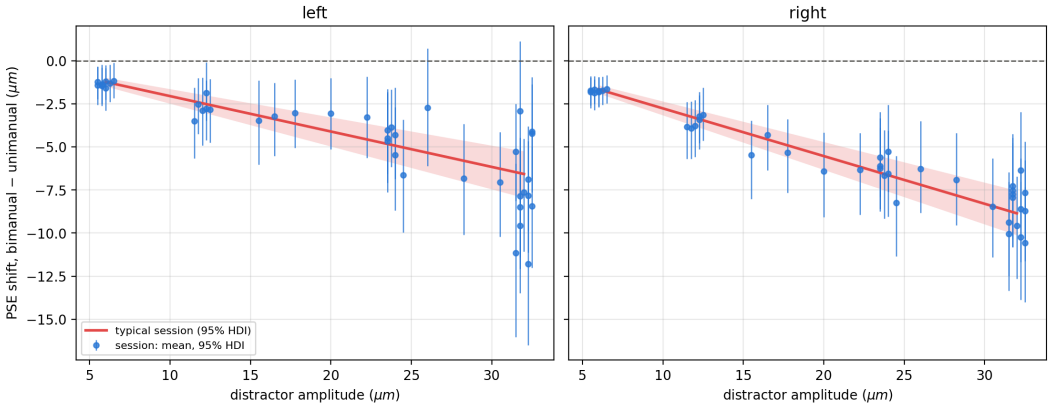
The bimanual curve (red) moves left: PSE about 4 μm lower (left hand), 6 μm (right)

The shift appears in nearly every session



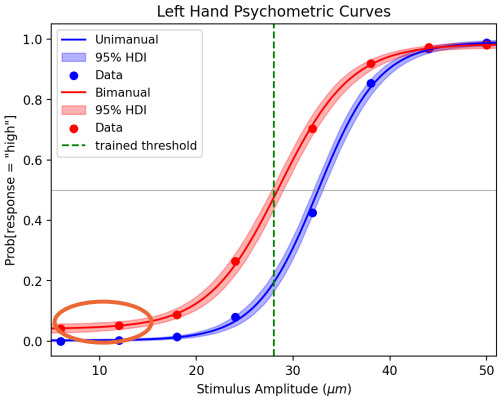
One session per point, in date order (Model B). Below 0: the distractor lowers the PSE

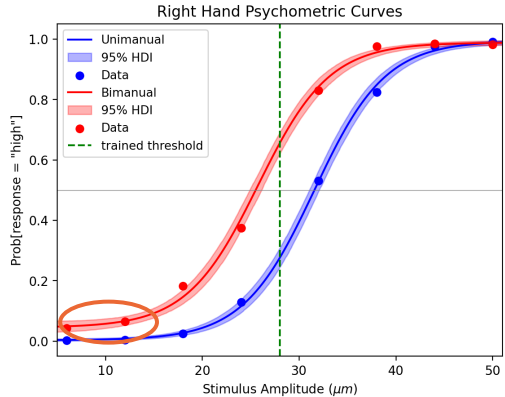
The shift grows with the distractor's amplitude



One session per point (Model C). But amplitude also rose over time

A floor under the bimanual curve





With a distractor, the monkey sometimes answers “high” to the weakest stimuli

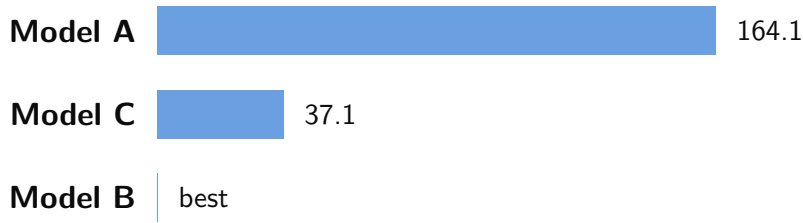
Less clear

- **Sensitivity:** the left-hand JND is slightly larger with a distractor
  driven by a few sessions; may be the same extra "high" answers
- **Weak distractors:** on the right hand, a 6 μm distractor shifts the PSE more than its amplitude predicts
  not yet shown to matter for prediction

Which model predicts best?

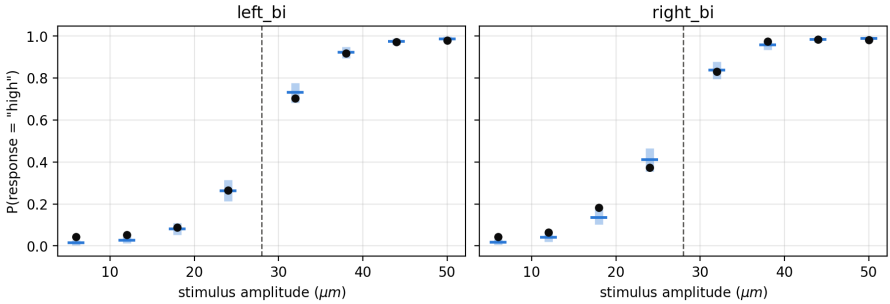
How much worse than Model B each model predicts left-out trials

(difference in log predictive score)



- B $\gg$ A: sessions really differ
- B $>$ C: C has one floor for both trial types

Why Model C falls short



Black: observed. Blue: Model C's prediction.

On bimanual trials, C predicts too few "high" answers to the weakest stimuli

Summary

- The distractor lowers the PSE: about 4 μm (left hand) and 6 μm (right), in all three models
- The shift is larger in sessions with a stronger distractor
- With a distractor, the monkey sometimes answers “high” to the weakest stimuli
- Sessions differ, so each session needs its own curve

Questions for you

1. **The extra "high" answers:** they seem to cluster in sessions with a strong distractor (being checked). Could the monkey sometimes respond to the distractor hand?

2. **Amplitude and time:** the distractor amplitude rose over the sessions, so a growing effect could also be training or adaptation. Was the schedule chosen for a reason? Did anything else change over those months?

Next steps

- **Model D:** separate lapse rates for unimanual and bimanual trials
- Test whether the bimanual lapse rate depends on the distractor amplitude, or on time
- Check Model D on simulated data before fitting the real data